Thermal radiation and color

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use peak wavelength as one piece of evidence about a thermal spectrum.

SUCCESS CRITERIA - I can...

1. Relate higher temperature to shorter peak wavelength.
2. Use kelvin in Wien's law.
3. Distinguish spectral peak from perceived color.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "blackbody" and "peak" to explain your model or result. Sentence starters are optional.

blackbody: an ideal absorber/emitter used in a thermal-radiation model

peak: the location of the largest value on a specified graph

kelvin: the absolute temperature scale used in thermal ratios

01 Warm-up: bring back an earlier idea

An object moves 0.3 m toward a fixed plane mirror. **Find** image movement and the change in object-image separation.

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: ① ② ③

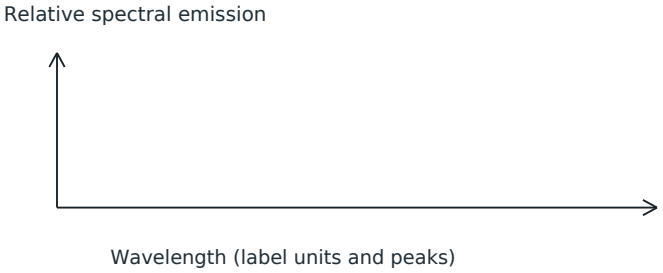
02 Worked example: explain the model

Reference: $\lambda_{max}\, T = 2.9\times10^{-3}$ m·K.
Ideal blackbody; T in kelvin, λ_{max} in meters. A shorter peak wavelength means higher temperature, not one exact visible color.

Worked reasoning: A thermal spectrum's peak shifts toward shorter wavelength as absolute temperature rises. Wien's relation $\lambda_{peak}\, T \approx 2.9\times10^{-3}$ m·K gives $\lambda_{peak} \approx 9.7$ micrometers at 300 K.

Your turn: Sketch two broad thermal spectra, labeling axes and the hotter peak's shift.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: ① ② ③

Thermal radiation and color

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two ideal thermal emitters are at 3000 and 6000 K. Which has the shorter peak wavelength? Does that mean they emit only one color?

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice … . This supports … because … .

Relative spectral emission



Wavelength (label units and peaks)

Marzano check - circle: ① ② ③

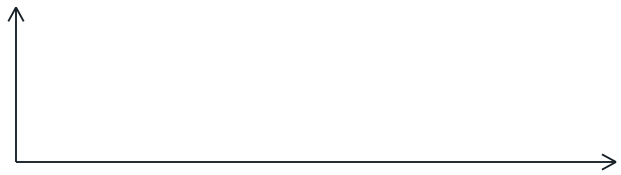
04 Apply the physics

Compare ideal thermal emitters at 5000 K and 10,000 K.
Use $\lambda_{peak}\, T = 2.9 \times 10^{-3}$ m·K to calculate peak wavelengths in nm.
Sketch both broad spectra.
Explain why their peaks alone cannot predict an exact observed meteor color.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both … . They differ in … because … .

Relative spectral emission



Wavelength (label units and peaks)

Marzano check - circle: ① ② ③

05 Explain your reasoning

An object has an infrared peak but looks red-hot.
Explain how both can be true.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

Across a stationary boundary, frequency remains 4 Hz while speed changes from 24 to 32 m/s.
Find both wavelengths.
Label a wave sketch to support the calculation or comparison.

Word help: boundary = the place where a medium or constraint changes

Sentence starter: I found … [unit]. This value means … in this situation.



Time or position: choose and label

Marzano check - circle: ① ② ③

Thermal radiation and color

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

An ideal blackbody has λ_{max}=725 nm.
Estimate temperature.

Word help: estimate = find an approximate value

Sentence starter: I found ... [unit]. This value means ... in this situation.

Relative spectral emission



Wavelength (label units and peaks)

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Find peak wavelengths at 3000 and 6000 K.
State why wavelength peak alone does not specify one exact perceived color.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Relative spectral emission



Wavelength (label units and peaks)

Marzano check - circle: 1 2 3